MILITARY SPECIFICATION

INTEGRATED VOICE COMMUNICATIONS SWITCHING SYSTEM

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance characteristics of an Integrated Voice Communications Switching System, hereinafter referred to as IVCSS, to be used primarily at Naval and Marine Corps facilities engaged in Air Traffic Control. The IVCSS shall provide many operators access to and control of multichannel radiophone, intercom and telephone communications. The IVCSS is the replacement for the current AN/FSA-52(V) and OA-7621(V)/FSA-52(V) Communications Control Groups.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specification, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-16400	Electronic Equipment, Naval Ship and Shore; General Specification
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of
MIL-P-116	Preservation-Packaging, Methods of
MIL-L-6363	Lamps, Incandescent, Aviation Service, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
DOD-STD-2167	System Software Development

PUBLICATIONS

NAVELEX 0967-874-5010	Technical Manual for Console Group, Communication Control AN/FSA-52(V)
NAVELEX 0967-005-9000	Technical Manual for Landline Selector Control Group OA-7261(V)/FSA-52(V) (Changes 1 & 2)
NAVELEX 0967-LP-005-9010	Technical Manual for Telephone Signal Converter, CV-2911A/FSA-52(V)
NAVELEX 0967-LP-005-9020	Technical Manual for Telephone Signal Converter, CV-2892A/FSA-52(V)
NAVELEX 0967-LP-005-9030	Technical Manual for Amplifier Signaling Unit, AM-6714/FSA-52(V)
NAVELEX 0969-LP-005-9040	Technical Manual for CV-3613/FSA-52 Telephone Signal Converter
EE165-EA-OMI-010/E120	Technical Manual for TA-1004/FSA-52(V) Telephone Signaling Unit
NAVELEX 0281-LP-071-1200	Technical Manual for AN/FSA-47 Transfer Group
EE165-EA-OMI-010/E120	Technical Manual for TA-1004/FSA-52 Telephone Signaling Unit
NAVELEX 0967-LP-429-5010	Technical Manual for AN/GRT-21,22
NAVELEX 0967-LP-428-1010	Technical Manual for AN/GRR-23,24
NAVELEX 0967-LP-544-5010	Technical Manual for AN/GRC-171,211

HANDBOOK

MILITARY

MIL-HDBK-241	Design Guide for EMI Reduction in Power Supplies

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2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536	Procedures and Requirements for Preparation in Soldering of Electrical Connections

FEDERAL COMMUNICATIONS COMMISSION

Part 68	Descriptions of Standard Registration Program Connection Configurations

(Copies of specifications, standards, handbook, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contractinglactivity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-310-C-77	Racks, Panels and Associated Equipment
RS-422-A	Electrical Characteristics of Balanced Voltage Digital Interface Circuits

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a

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sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 IVCSS description. The IVCSS shall be a digital non-blocking distributed microprocessor controlled system which provides the means for establishing communications control at Navy and Marine Corps ATC and Range facilities where it is installed. The various types of multi-channel communications include radiophone, intercom and telephone.

3.2.1 Radiophone. This portion of IVCSS shall provide air-ground communications between operators and pilots in aircraft. The transmission medium shall be radio, and IVCSS shall be required to interface with both local and remote Government furnished transmitters and receivers for this purpose.

3.2.2 Intercom and telephone. The intercom portion of the IVCSS shall enable operators to talk to other operators at the same facility while the telephone portion allows operators to communicate with operators at remote locations. The intercom and telephone portions are usually referred to as ground-ground communications.

3.2.3 Major subsystems. The various subsystems of the IVCSS shall be as specified in a through g:

a. Operator control modular equipment subsystem.

b. Control central equipment subsystem.

c. Configuration control equipment subsystem.

d. Supervisor control equipment subsystem

e. Maintenance diagnostic support equipment subsystem.

f. Maintenance control equipment subsystem.

g. Power equipment subsystem

3.2.4 Interrelationship of subsystems. Figure 1 illustrates the interrelationship of the various subsystems which comprise the IVCSS.

3.3 Characteristics. Characteristics shall be as specified in 3.3.1 through 3.3.9.

3.3.1 General characteristics. The IVCSS shall provide at each operator position in a facility the ability to initiate or receive voice communications with other local operators, operators at other facilities, pilots, or any combination thereof. Although the IVCSS will be procured in specific configurations, the operational requirements may vary from facility to facility. Therefore, the basic building blocks of the IVCSS shall be modular in nature, so that different hardware configurations can be assembled from a minimum number of distinct elements. The overall system shall have the capacity (non-wired) to provide for a minimum of 104 operator positions, 124 radio channels and 132 intercom/telephone channels. Specific system configuration and capacity shall be specified in the contract (see 6.2.1). A

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configuration control capability shall be provided which will allow the operational features and communications assignments of all positions to be reconfigured electronically.

3.3.1.1 Operator control subsystem. Each operator control subsystem, which is commonly referred to as the operator position, shall consist of the following:

a. Radiophone switch modules

b. Intercom/telephone direct access switch modules

c. Selector display modules

d. Special function module

e. Speaker control module(s)

f. Dimmer module

g. Jackbox module

h. Headset level control module

i. Footswitch

j. Position processor module

k. Headset

3.3.1.2 Control central subsystem. One control central subsystem shall consist of the following:

a. Switching network

b. Position interface

c. Radio interface

d. Telephone interface

e. Recorder interface

f. Power distribution

g. Supervisor interface

h. Maintenance interface

3.3.1.3 Configuration control subsystem. This subsystem shall consist of an interactive terminal, associated hardware and software. The terminal is located at the supervisor position.

3.3.1.4 Supervisor control subsystem. This subsystem, which is commonly

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referred to as the supervisor position, shall consist of the following:

a. Multi-page touchscreen entry/display for all radio, intercom and telephone communications

b. Speaker control module

c. Jackbox module

d. Footswitch

e. Supervisor position processor module

f. Main/standby transfer control module(s)

g. Interactive terminal

h. Handset

3.3.1.5 Maintenance diagnostic support subsystem. This subsystem shall consist of an interactive terminal, associated hardware and software. The terminal is located in the maintenance area adjacent to the maintenance position.

3.3.1.6 Maintenance control subsystem. This subsystem, which is commonly referred to as the maintenance position, shall consist of the following:

a. Multi-page touchscreen entry/display for all radio, intercom and telephone communications

b. Speaker control module

c. Jackbox module

d. Maintenance position processor module.

e. Handset

3.3.1.7 Power equipment subsystem. This subsystem shall consist of redundant DC power supplies with complete battery back-up capability.

3.3.2 Specific subsystem characteristics

3.3.2.1 Operator control subsystem. This subsystem provides the operator with necessary equipment to initiate or respond to radiophone, intercom and telephone communications. The specific requirements for an operator position are as specified in 3.3.2.1.1 through 3.3.2.1.12. Space availability for the various modules shall be as depicted on Figure 2.

3.3.2.1.1 Radiophone switch modules (RSM). Each RSM shall provide access to a minimum of four radiophone channels. All switches shall have appropriate indicator lights to provide status of each function. An option shall be provided on an individual channel basis for lockout on selection or lockout on key. The lockout option shall be electronically alterable. The module shall

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operate in conjunction with alphanumeric displays adjacent to each radio transmit select pushbutton which can be electronically alterable to correspond to the current IVCSS configuration. Alternate labeling capabilities shall be provided such as removable designator strips adjacent to the switches or removable legends within the pushbutton housing. The designator strips and pushbuttons shall be backlighted under the control of the common dimmer control. As a minimum, the following switch functions shall be provided for each radiophone channel.

a. RSM transmit select pushbutton. When the RSM transmit select pushbutton is depressed, the particular channel is selected by the operator for transmitter keying and receiver monitoring. This selection shall be indicated by energizing a green status lamp. Depressing the pushbutton again will release the channel. If a channel configured for lockout on select has been selected for transmitting at another operator position, a red status light will be energized. If a channel configured for lockout on key is selected for transmitting at another operator position, the red status light will be off until the channel is keyed. When the channel is keyed, the red status light will flutter until the key (PTT) is released. When the channel is not selected by any operator, the green and red status indicators will be off and the channel shall be available for selection.

b. RMS receiver monitor switch. When the RSM receiver monitor switch is depressed, the particular channel's receive audio will be monitored in the headset or speaker mode. This switch also determines the receive mode (headset or speaker) of the RSM transmit select pushbutton. The switch must provide for three selection modes: voice shall be either off, in the headset, or in the loudspeaker. Two status lights shall display the selected mode. These lights shall also provide the VOX indication (flutter) for presence of receive radio audio. When a channel has been selected for transmission by the RSM transmit select pushbutton and the receiver monitor switch is in the off position, the receive audio shall be directed to the headset. When specified in the contract (see 6.2.1), some quantities of an alternate RSM configuration shall be required. In addition to the characteristics specified above, the alternate RSM shall allow the operator to select between main and standby radio transmitters and receivers on an individual channel basis. This shall require two additional pushbuttons per radiophone channel as described below. Backlighting under the control of the common dimmer control shall be provided.

c. RSM transmit main/standby pushbutton switch. This pushbutton switch provides the operator with the capability to switch between main and standby radio transmitters. Status lights shall be provided to indicate whether the main transmitter (green) or standby transmitter (amber) is selected. Selection is accomplished by depressing the switch.

d. RSM receiver main/standby pushbutton switch. This pushbutton switch provides the operator with the capability to switch between main and standby radio receivers. Status lights shall be provided to indicate whether the main receiver (green) or standby receiver (amber) is selected.

3.3.2.1.2 Selector display modules (SDM). Each SDM shall contain a number of six-character alphanumeric word displays equal to the number of selectors in the selector module. Decimal points, for example, between the third and fourth digits of a radio frequency (RF) display, shall not be

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counted as one of the six required characters. Each display module shall be located in or adjacent to a selector module. The 26 upper case letters "A" to "Z" and the ten numerals "0" to "9" shall be displayed in distinct and unambiguous fashion. Display module words shall be legible in a broad range of ambient light, ranging from total darkness to bright light (not less than 40 foot candles per square foot). Variable intensity lightning and backlighting shall be provided, as appropriate, under control of the common dimmer control.

3.3.2.1.3 Intercom/telephone direct access switch module (DASM). Each operator control subsystem shall have the capability to directly access intercom/telephone channels. These channels provide direct access to operator positions within the same facility or operator positions at remote facilities. The module shall contain a minimum of four pushbuttons and shall be expandable in groups of four pushbuttons to a total of sixteen. One pushbutton switch with appropriate indicator light(s) shall be provided for each direct access intercom/telephone channel. Indicator lamps shall provide the operator with instantaneous recognition of pushbutton assignments (i.e.: intercom call, intercom monitor, telephone). Pushbutton labeling shall be provided by removable designation strips adjacent to the pushbuttons or by removable legends within the pushbutton housing. The module shall have the capability of accommodating alphanumeric displays (see 3.3.2.1.2) adjacent to each direct access pushbutton which can be electronically alterable to correspond to the current IVCSS configuration. The DASM shall provide for three distinct modes of intercom operation; intercom call, intercom call with override, and intercom monitor. Pushbuttons and designation strips shall be backlighted under control of the common dimmer control.

3.3.2.1.4 Special function module (SFM). Each operator control subsystem shall have a special function module which includes a pushbutton dial keypad and the following additional pushbuttons/indicators:

Indirect access (IA)
Call answer (CA)
Hold (HLD)
Release (REL)
Headset/loudspeaker (HL)
Override headset/loudspeaker (OHL)
Programmable (Qty 2)

Each SFM shall contain not less than of 8 pushbuttons (in addition to dial keypad buttons). Each pushbutton (with the exception of the hold and release) shall have a designation strip/legend and status light. Selection of each pushbutton shall energize the associated status light. All pushbuttons shall be momentary type. Backlighting of all pushbuttons shall be under the control of the common dimmer control. The requirements for this module shall be as specified below:

a. Pushbutton dial keypad. Provides capability to selectively signal on intercom/telephone channels, to initiate self test and to enter various access codes. Provides both standard dual tone multi-frequency (DTMF) and dial pulse signaling capabilities.

b. Indirect access (IA) pushbutton. When used in conjunction with the pushbutton dial keypad, the indirect button shall be used to initiate intercom and telephone calls which are not otherwise accessible through a DA pushbutton. All DA accessible parties shall also be accessible through use of

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the IA procedure. An indirect access shall be initiated by an initial operation of the IA pushbutton followed by the appropriate address code entered on the keypad. A self test of the position shall be provided and activated by an indirect access self test code. This feature shall execute test routines in the position processor to perform diagnostics verifying operational status of the position. Appropriate pass or fail indications shall be provided. In addition, a means for testing all indicator lights shall be included. A reconfigure enable function shall also be provided which allows an operator to control the timing of the reconfiguration of his position. After system reconfiguration has been initiated by the supervisor, the operator position shall receive a visual and audible indication that the position is ready for reconfiguration. However, reconfiguration can not occur until the operator executes an indirect access code on the SFM keypad. An audible indication shall then be received at the position indicating that position reconfiguration is complete. A self reconfiguration function shall also be provided which shall allow each operator to select not less than two of the pre-programmed position maps described in 3.3.2.3.1. The reconfiguration shall be initiated by entering an indirect access code on the SFM keyboard. No other positions shall be affected by this self reconfiguration. The access code shall be changeable.

c. Call answer (CA) pushbutton. The call answer pushbutton shall provide the capability to answer all calls not directed to DA buttons and provide a visual indication.

d. Hold (HLD) pushbutton. The hold pushbutton shall establish a hold condition on all intercom (IA) and telephone calls, except override calls, to enable the operator to answer or initiate another call without disconnecting the call on hold. It shall be possible to establish a hold condition on multiple circuits at a position simultaneously. The indicator light of each circuit on hold shall wink. A connection shall be reestablished by depressing the button associated with a circuit on hold thereby removing the hold condition for that circuit.

e. Release (REL) pushbutton. The release pushbutton shall disconnect the position from any existing intercom or telephone connection except an incoming override call. The release button shall have no status indicators.

f. Headset/loudspeaker (HL) pushbutton. Successive operation of the HL button shall direct the audio routing of incoming intercom/telephone and radio calls either to the headset or to the loudspeaker as chosen by the operator.

g. Override headset/loudspeaker (OHL) pushbutton. Successive depressions of the OHL pushbuttons shall direct the routing of incoming override calls either to the headset or the loudspeaker as chosen by the operator, unless the HL switch is in the speaker position. In this case, the HL switch shall govern the routing of incoming override calls. Also provides a visual indication of an incoming override call.

i. Programmable pushbuttons. Two programmable pushbuttons shall be provided on the SFM which may be programmed to provide any one of the functions specified in 3.3.2.1.11.

3.3.2.1.5 Speaker control module (SCM). The SCM provides a loudspeaker

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mounted behind a grill for reproducing voice and chime signals. Front panel mounted level controls, accessible to the operator, shall be provided to adjust the audio levels to the headset and loudspeaker. In addition, the SCM shall provide necessary circuitry to generate chime signals. The chime signal shall be generated at approximately two second intervals whenever an intercom (if configured for chime) or telephone call is directed to the position. The chime shall cease when the call is answered. Maintenance controls, not accessible to the operator, shall be provided for adjusting the chime volume and frequency. The maximum chime output level shall be limited to 300 mw into a 4 ohm resistive load. A front panel mounted chime cut-off pushbutton shall silence the chime tone. Successive operation of the button shall alternately mute the chime and allow incoming chimes to ring. The status light associated with the chime cut-off pushbutton shall be on only when the chime is muted. Two SCM's are required at positions designated for split operation.

3.3.2.1.6 Dimmer module. A common dimmer control shall be provided to vary the intensity of all backlighted pushbuttons and indicators at the position. It may be a separate module or included on the SFM or SCM.

3.3.2.1.7 Jackbox module. This module shall provide a multifunction interface between the system and the operators. It shall contain a dual set of tip-ring-sleeve jacks for use with 6-wire headsets/handsets equipped with a PJ-421 plug. The jacks shall occupy minimal space in a front-panel mounted module but shall be sufficiently separated to allow headsets and handsets to be easily inserted and removed without undue stress on the jackbox or headset/handset. One set of jacks shall be clearly designated for use by the operator and the other set shall be clearly designated for monitoring. For clarification purposes, the operator plugged into the operator jacks will be referred to as the main operator while the operator plugged into the monitor jacks will be referred to as the monitor operator.

a. Normal operating mode. In this mode, the main operator shall be able to communicate on intercom and telephone channels without depressing his headset's PTT switch. To communicate on radio, the operator must depress the PTT switch or the position's footswitch. The monitor operator will only be able to monitor all communications unless the following actions are taken. To communicate on intercom and telephone, the monitor operator must depress his headset/handset's PTT switch. To communicate on radio, the monitor operator must depress the PTT switch and the position's footswitch.

b. Split operation mode. If the Split Operation (SO) button on the SFM is selected, the main operator shall have access to radio communications only, while the monitor operator shall have access to intercom/telephone communications and shall be able to hear radio communications except when intercom/telephone communications is in progress. The monitor operator will not have the capability to communicate over a radio channel while operating in the SO mode.

c. Instructor mode. If the Instructor (INST) pushbutton on the SFM or DASM is selected, the monitor operator shall have pre-emptive capabilities over the main operator regardless of whether the operator is transmitting on a radio, intercom or telephone channel. The monitor operator initiates pre-emption by depressing the headset/handset's PTT switch. When not pre-empting, the monitor operator will be able to monitor all radiophone,

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intercom and telephone communications. The monitor operator can only communicate if the PTT switch is depressed. When pre-empted, the main operator can only monitor radiophone, intercom and telephone communications. When specified in the contract (see 6.2.1), some quantities of separate modules containing the monitor operator jacks shall be provided. This will allow for a more convenient location when operating in the SO or INST modes.

3.3.2.1.8 Headset level control module (HLCM). A separate level control shall be provided for the monitor operator jacks when in the split operation or instructor pre-empt mode. The level control may be contained in a separate module or included in another operator position module.

3.3.2.1.9 Footswitch. A moveable footswitch which is both functionally and physically equal to the SA-585/FSA-17 (P/O AN/FSA-52) shall be provided at each operator position. It shall be connected in parallel with the PTT switch of the 6-wire headset/handset. A panel connector shall be used to terminate a footswitch cable at each position. The cable shall not interfere with the operator.

3.3.2.1.10 Operator position processor module (OPPM). The modules described in 3.3.2.1.1 thru 3.3.2.1.9 shall be accessible to the operator. All other circuitry necessary for interfacing the position modules to control central may be rack mounted in the existing operator console cabinet in a location not accessible to the operator. However, the equipment shall be readily accessible to maintenance personnel. This module includes audio and control processing circuitry for radiophone, intercom and telephone communications, the self-testing function, the reconfiguration function, position power supplies (if required), and interconnecting cable terminations. The OPPM shall be capable of interfacing touch entry display devices as well as conventional pushbutton modules. Each position processor shall communicate with the appropriate processor at control central via an industry standard dual bus network such as RS-422. The dual bus arrangement shall be required to minimize single point failures which could disrupt vital communications.

3.3.2.1.11 Headset. A lightweight headset (Plantronics MS50T62-30-2 with background noise suppressor P/N 05354-00) or equal, shall be provided with each operator position.

3.3.2.1.12 Programmable feature capability. Each operator control subsystem shall have access to the following special features on a programmable basis. Access shall be via the programmable pushbuttons located on the SFM or any of the pushbuttons located on the DA module.

a. Manual ring. This feature shall allow the operator to manually initiate ringdown telephone calls.

b. Instructor pre-empt. This feature shall enable an instructor to pre-empt an operator on radio, intercom and telephone channels. Selection of the INST button shall automatically release the SO button (see 3.3.2.1.7.c.).

c. Call forwarding. The system shall permit any operator to forward incoming intercom and telephone calls to any other local operator.

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d. Call transfer. This feature shall enable any operator to transfer any incoming intercom or telephone call, except override, to any other operator within the facility.

e. Door release. This feature shall enable any operator to release electrically activated facility doors.

f. Conference. This feature shall provide the capability to establish conference intercom calls connecting up to fifteen operators. The received audio level at each position shall be independent of the number of positions connected in conference.

g. Speed-dialing. This feature shall provide an operator with the capability to speed-dial on PBX, SS-1A/SS-4 and local type telephone circuits. Pushing a direct-access button at the position shall access the appropriate circuit and generate the necessary signaling (up to ten digits) to call the remote party.

h. Split operation (SO). This feature shall permit two operators to simultaneously use one position. One operator shall use intercom/telephone communications, and shall also be able to hear all radio communications except when intercom/telephone communications are in progress, while the other operator shall use only radio communications.

3.3.2.2 Control central subsystem. This subsystem provides the switching network, power distribution system and the various interfaces to position, radio, telephone, maintenance and supervisor equipment. If remote central equipment is necessary to meet the position distance requirements of 3.3.2.2.2, it shall have the same characteristics as the main central subsystem. The remote central equipment shall act as a slave of the main central subsystem. Reconfiguration of the remote central equipment and positions shall be possible from the main central subsystem.

3.3.2.2.1 Switching network. This equipment shall provide the means for routing all radio, intercom and telephone communications. The switching network shall be under the control of software driven distributed microprocessors. These microprocessors shall be contained in individual position, radio and telephone elements of the IVCSS.

3.3.2.2.2 Position interface. This equipment provides the necessary interface between the switching network and the position processor modules located at each position. It shall contain necessary circuitry to process all voice communication signals (radio, intercom, telephone) control signals and status signals between the control central subsystem and the operator control subsystem. Maximum distance between the control central subsystem and the operator control subsystem shall be 400 feet without the need for additional position power supplies. The maximum distance shall be extended to 4000 feet with the use of additional position power supplies. If specified in the contract (see 6.2.1), the position interface shall provide the capability to have remote positions connected via privately owned or commercial networks. Distances between the IVCSS control central and the remote position(s) shall be unlimited. Each position shall require two duplex circuits for voice and control.

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3.3.2.2.3 Radio interface. This equipment shall provide the necessary interface between the switching network and the Government furnished radios located at either local and remote radio sites. It shall contain necessary circuitry to process radio VF and control signals. This includes VF amplification, impedance matching, keying, interlock, status and protection devices for both main and standby channels. Local GFM radio sites will be located in the proximity of the air traffic control (ATC) facility (10,000 feet max.). Specific radios include the AN/GRR-23,24 VHF/UHF receivers, AN/GRT-21,22 VHF/UHF transmitter, AN/GRC-171 UHF transceiver and AN/GRC-211 VHF transceiver. Jack panels utilizing miniature (switchcraft TT-series, ADC Bantam or equal) type jacks shall be provided to allow easy access to each radiophone channel (transmit, receive and key) for maintenance and radio patching purposes. The jacks shall allow maintenance personnel to connect test equipment to the line side, equipment side or to bridge the circuit under test. The interface will be required to provide the following connections.

a. One pair for the VF signal to the main transmitter.

b. One pair for the VF signal to the standby transmitter.

c. One pair for the VF signal from the main receiver.

d. One pair for the VF signal from the standby receiver.

e. One pair for keying for the main transmitter (dry contact, 100 ma maximum).

f. One pair for keying for the standby transmitter (dry contact, 100 ma maximum).

g. Provision for connecting centertap of transmit output transformer to ground or voltage (50 VDC max) for simplex keying. Maximum simplex current shall be 50 ma. When specified in the contract (see 6.2.1), the radio interface shall also include in-band tone signaling for control of remote transmitters and receivers connected via trunk lines.

h. When specified in the contract (see 6.2.1), the radio interface shall provide the capability to connect the IVCSS to a variety of FM VHF transceivers in use at Navy facilities.

3.3.2.2.4 Telephone interface. This equipment shall provide the necessary interface between the switching network and telephone systems and equipment external to the IVCSS on both a two and four wire basis. It shall contain necessary circuity to process telephone VF and control signals for a multiplicity of telephone circuits normally encountered at Navy ATC facilities. This includes VF amplification, impedance matching, 4-wire to 2-wire conversion, signaling (outgoing and incoming) and protection devices. All termination equipment shall be in compliance with part 68 of the Federal Communications Commission (FCC) rules. A registration, certification number and ringer equivalence shall be obtained. Jack panels utilizing miniature type jacks shall be provided to allow easy access (line side, equipment side and bridging) to each telephone channel for maintenance and patching purposes. All telephone interface equipment shall be modular and compatible with the following circuit configurations and signaling methods. A listing by

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Navy circuit types (including existing Navy OA-7621/FSA-52 termination equipment) is provided in Table 1.

a. Voice call circuit (Navy Types 1 and 5). Shall provide high priority communications between ATC facilities and a private voice network. A call is initiated by selecting the appropriate DA pushbutton and aurally calling the called party at the distant facility. The aural signal is heard over a programmed speaker at the distant facility. To reply, the called party simply "selects" the circuit. After selection, the calling party's voice is transferred from speaker to the headset of the called position. Likewise, calls from distant facilities are heard over the programmed position's speaker in the IVCSS and are transferred to the headset upon selection of appropriate DA button.

b. Manual ringdown circuit (Navy Types 2, 3, 4, 6, 8, 8A, 8B, and 9). Nonselective private line circuit used for point-to-point communications between ATC facilities. A call is initiated by selecting the appropriate DA pushbutton and then manually signaling (by means of Ring pushbutton) or in some cases, by voice, depending upon the type of equipment at the distant facility. Incoming signaling is Ring or in some cases, voice. Both outgoing and incoming ring signaling may be 20 Hz, 105VAC or 2600 Hz single frequency (SF).

c. Automatic ringdown circuit (Navy Type 16). Nonselective private line circuit used for point-to-point communications between ATC facilities. Outgoing signaling is automatic upon selection of programmed DA pushbutton. Incoming signaling shall be ring (20 Hz, 105VAC).

d. Selective signaling circuit (Navy Types 7, 10 and 11). Shall provide selective signaling (send and receive modes) over 4-wire multi-party private line voice networks. A call is initiated by first selecting the appropriate DA pushbutton and then dialing a 2 or 3 digit code assigned to the called position at a distant facility or a programmed position within the IVCSS. Selective signaling equipment shall be compatible with Western Electric SS-1, SS-1A and SS-4 systems. Type 10 circuits utilize selective signaling in the send mode only. Incoming signaling is voice via the position speaker.

e. Central office (CO) and PBX extension circuits (Navy Type 12). Shall provide dial selective outgoing (pulse or DTMF) and nonselective (20 Hz Ring) incoming signaling over a two-wire line which is connected directly to a CO or to a local PBX.

f. Local single station circuit (Navy Type 18). Shall provide private two way communication between positions within the IVCSS and local vicinity points (such as weather or line shack) over a 2-wire line. Outgoing signaling from the IVCSS may be initiated automatically (selection of DA pushbutton only), manually initiated or fully manual. Local vicinity points utilize a standard common battery telephone instrument (W. E. Type 2500 or equal). Outgoing calls from the remote instrument may be initiated automatically (removal of telephone instrument's handset) or selectively (not less than 10 codes) via the DTMF touchpad.

g. Local multi-station multi-link circuit (LMSML). Shall provide communications with selective dialing capabilities between operator positions

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within the IVCSS and remote positions (up to 16) equipped with standard 2-wire single line telephone instruments. In addition, the equipment shall allow each of the remote positions to selectively signal and communicate with each other and up to six positions within the IVCSS. Conferencing capabilities as well as a minimum of two talk links shall be provided. Other specific requirements shall be as specified in the TA-1004/FSA-52 technical manual.

h. Dial pulse signaling. The IVCSS shall be capable of generating dial pulse signals. The necessary circuitry may be located at the position or in the telephone interface equipment. Outgoing dial pulses shall be transmitted at 8.5 to 10.5 pulses per second with a 58 to 62 percent break. The interdigital time between the last pulse of one digit and the first pulse of the next digit shall be not less than 600 ms.

i. Ringdown signaling. The IVCSS shall be capable of receiving and detecting 20 Hz, 55-105VAC signaling and transmitting 20 Hz, 105VAC signaling.

j. DTMF signaling. The IVCSS shall be capable of transmitting and receiving DTMF signals. Table II specifies the pushbutton digits, designations and corresponding tone frequencies.

k. E & M signaling. When specified in the contract (see 6.2.1), the IVCSS shall provide the capability to use (directly) E & M signaling on facilities leased from the common carrier or provided by the Government. DC conductor resistance in the E lead and M lead shall be limited to 25 ohms or less. The following convention shall be used on the E & M signal or the line side of the E & M signaling module. When the circuit state is "on-hook", the E-lead shall be open and the M-lead grounded. When the circuit is "off-hook", the E-lead shall be grounded and the M-lead shall be -48 VDC.

l. SF signaling. The IVCSS shall provide for transmission of SF signals over two-wire and four-wire VF channels without interfering with their use for speech.

m. Beep tone. The IVCSS shall provide a beep tone generator with multiple outputs capable of supplying beep tones to PBX circuits for recording purposes. Table III specifies the level and frequency requirements.

n. Administrative phone system (APS). If specified in the contract (see 6.2.1), an administrative phone system (APS) shall be provided. The APS shall be a stand alone electronic type key system and shall provide access to central office (CO) trunks with access to the IVCSS via tie-lines. In addition, an intercom feature shall be provided allowing for selective communications between stations. Stations shall consist of either single line or multi-line smart sets (or a combination of both). The APS shall be provided in the following configurations:

CO Trunks	Stations
28	64
8	16

o. PBX tie line. Two-wire circuit that provides selective incoming (two and three digit DTMF receive) and outgoing (pulse or DTMF) signaling. Shall

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be capable of interfacing with a CO line, PBX extension or a local facility administrative phone system.

3.3.2.2.5 Voice recording interface. This equipment shall provide necessary facilities for combining all incoming and outgoing radio, intercom and telephone communications into a single signal for position recording. This signal shall at all times represent the relative level of what the position operator hears. When a position is operated in the split operation (SO) mode, separate recording paths shall be provided. In addition, facilities shall be provided for recording all radio frequencies and telephone circuits. Both position and channel record signals shall be normalized to provide a nominal OdBm signal to the GFE recorder. Jack panels utilizing miniature type jacks shall be provided to allow easy access to each position record channel for maintenance and patching purposes.

3.3.2.2.6 Power distribution and protection. The IVCSS shall provide necessary facilities for primary/secondary power distribution and fuse protection.

a. Primary power protection. Fusing shall be provided for each power source. In addition, means shall be included to allow all power sources to be easily disconnected from the main power busses.

b. Secondary power protection. Fused circuit protection with blown fuse indicating circuitry shall be provided for all control central and position equipment subsystems. Control central equipment shall be fused on an individual channel basis. Each position's electronic and illumination circuits shall be fused separately. Blown fuse indications (faults) shall be reported through the maintenance diagnostic equipment subsystem. However, visual indication of which fuse has blown shall be provided on the associated fuse panel.

3.3.2.2.7 Supervisor interface. This equipment shall be identical with the position interface described in 3.3.2.2.2 except for those changes necessary to accomodate any operational differences which may exist between operator and supervisor positions.

3.3.2.2.8 Maintenance interface. This equipment shall be identical with the position interface described in 3.3.2.2.2 except for those changes necessary to accomodate any operational differences which may exist between operator and maintenance positions.

3.3.2.3 Configuration control subsystem. The IVCSS configuration control equipment shall provide the capability to reconfigure the operational features and communications assignments of all operator positions.

3.3.2.3.1 Configuration maps. Reconfiguration shall be accomplished primarily through the use of pre-programmed instructions ("maps") in both a global and local mode. The maps shall be contained in a reconfiguration database. In the global mode, assignments for the entire facility are made, while in the local mode, individual operator position assignments are made. The reconfiguration maps provide the following functions and classmarks:

a. Assignment of intercom and telephone channels at all positions.

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b. Assignment of intercom (non-override) call mode (voice or chime)

c. Assignment of radio channels at all positions.

d. Assignment of main/standby channels.

e. Assignment of special function pushbuttons at all positions.

f. Assignment of alphanumeric messages in position display modules.

g. Assignment of locking and non-locking pushbuttons at all positions.

h. Assignment of alternate method of terminating intercom calls.

i. Assignment of radio channel lockout modes (key-to-lockout or select-to-lockout).

j. Assignment of radio channel preemption.

k. Assignment of radio channel receiver monitoring (PTT restricted)

l. Assignment of received radio muting modes.

m. Assignment of DA and IA override

n. Assignment of telephone signaling to specific operator positions.

o. Reconfiguration initiation authorization

p. Capability to accomplish all of the above assignments when adding or deleting positions.

3.3.2.3.2 Reconfiguration method. Reconfiguration shall be semi-automated and under the control of a designated operator using the interactive terminal located at the supervisor position. Prior to the implementation of reconfiguration, the authority, as classmarked, of the supervisor position requesting the reconfiguration shall be validated. When a reconfiguration is pending, visible and audible indications that the reconfiguration is ready shall be provided at each affected operator position which has a reconfigure enable capability. The position shall then reconfigure only when the position operator executes the appropriate indirect access code. The position shall then give an audible indication that the position reconfiguration is complete. At each position not having a reconfigure enable capability, the position shall be automatically reconfigured when all connections at the position have been broken. The position shall receive no indication of pending reconfiguration but shall receive an indication of completed reconfiguration. Reconfiguration of a single operator position (local reconfiguration) and reconfiguration of the entire IVCSS (global reconfiguration) shall be accomplished within one minute. Nonvolatile storage devices (EPROM) shall be incorporated into the various system processors. The function of these devices shall be to retain configuration data upon loss of power. Upon reapplication of power, the system shall re-boot automatically with the current map assignment.

3.3.2.3.3 Interactive terminal control. The interactive terminal

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provided at the supervisor position shall allow a designated operator to select and execute reconfigurations, either from available configuration maps or as defined by the operator. The terminal equipment shall include a keyboard, data display, non-sequential bulk storage device, printer and necessary control hardware and software. A screen dump of the terminal, which includes system status display, system units editing, local/global maps assigned, shall be printed on demand. The interactive terminal shall display the status of the reconfiguration while it is in progress. This shall include a list of positions that have not been enabled and those that have been reconfigured, together with the time of reconfiguration. A reconfiguration message shall be transmitted to the maintenance terminal printer when the reconfiguration has been completed. The supplied software shall allow the designated operator to create, update and maintain a reconfiguration database. In addition, the software shall allow access to the reconfiguration database to be restricted (password required). The capability shall be provided to define, at a minimum, 5 position maps per position in the reconfiguration database. Creation and maintenance of the database shall be accomplished in an off-line mode. If specified in the contract, a second interactive terminal shall be provided at joint Navy facilities (i.e., Navy ATC and FAA, Navy ATC and Navy Ranges).

3.3.2.4 Supervisor control equipment subsystem. This subsystem provides a designated operator with necessary equipment to communicate on radiophone, intercom and telephone channels through the utilization of a multi-page touchscreen entry/display. The specific requirements for the supervisor position shall be specified in 3.3.2.4.1 through 3.3.2.4.7.

3.3.2.4.1 Multi-page touchscreen entry/display module (MPTED). This module shall be functionally equivalent to the RSM, DASM and SFM. It shall consist of a programmable high resolution cathode ray tube (CRT) color display with a minimum of 640 x 200 pixels and be capable of producing not less than 8 colors with highlight and blink attributes. The size of the display shall be not less than 9 inches (measured diagonally) and not greater than 14 inches. The screen display shall be easily tailored to accomodate a variety of operational requirements. The CRT shall be flat in at least one dimension to minimize reflections. It shall employ touch detection techniques that provide not less than 1000 x 1000 touch points without compromising display brightness and contrast. In lieu of the CRT, a programmable touch panel using a light emitting diode (LED) based dot matrix display or a vacuum fluorescent (VFL) display capable of producing both alphanumerics and graphics may be utilized. The LED and VFL displays shall be sunlight readable and possess a conical viewing angle of at least 120 degrees. Infrared (IR) detection techniques shall be utilized with both the LED and VFL display panels. Page layouts for all displays shall provide a reasonable facsimile of the operator's RSM, DASM and SFM. Final layout shall be approved by the procuring activity.

The designated operator shall be able to perform the following functions:

a. Access all radio channels.

b. Monitor any operator via intercom monitor DA route.

c. Access all telephone channels.

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d. Access all other positions (operator and maintenance) via intercom DA and IA route.

e. Adjust brightness level of display.

3.3.2.4.2 Speaker control module (SCM). This module shall be identical with the SCM described in 3.3.2.1.5.

3.3.2.4.3 Jackbox module. This module shall provide the interface between the system and the designated operator using the supervisor position. It shall contain one set of tip-ring-sleeve jacks for use with 6-wire headsets/handsets equipped with PJ-421 plugs. The jacks shall occupy minimal space in a front-panel mounted module and shall be clearly designated for use by the operator.

3.3.2.4.4 Footswitch. This unit is identical with the footswitch described in 3.3.2.1.9.

3.3.2.4.5 Supervisor position processor module (SPPM). This module shall be identical with the OPPM described in 3.3.2.1.10 except for those changes necessary to accomodate any operational differences which may exist between operator and supervisor positions.

3.3.2.4.6 Main/standby transfer control module (TCM). This module provides the capability of selecting between main and standby radio transmitters and receivers on an individual channel basis and shall be functionally equivalent to the SB-1996/FSA-47. Operational requirements may dictate the need for TCM's to be installed in two distinct locations such as the Instrument Flight Rules (IFR) and Visual Flight Rules (VFR) areas of an ATC facility. Therefore, the main/standby transfer equipment must be modular to provide necessary installation flexibility. Each module shall contain not less than 4 channels and not greater than 8. Total TCM capability in each IVCSS shall be not less than 16 channels unless specified otherwise in the contract (see 6.2.1). Pushbutton labeling shall be by removable designation strips adjacent to the pushbuttons or by removable legends within the pushbuttons. Variable intensity backlighting shall be provided. The following main/standby transfer switch functions shall be provided.

a. Transmit main/standby pushbutton switch. This locking pushbutton switch provides an operator with the capability to switch between main and standby radio transmitters. Status lights shall be provided to indicate whether the main transmitter (green) or standby transmitter (amber) is selected.

b. Receiver main/standby pushbutton switch. This locking pushbutton switch provides an operator with the capability to switch between main and standby radio receivers. Status lights shall be provided to indicate whether the main receiver (green) or standby receiver (amber) is selected.

3.3.2.4.7 Handset. A 6-wire handset (W.E. G5JRW-3 or equal) shall be provided with each supervisor position.

3.3.2.5 Maintenance diagnostic equipment subsystem. This subsystem provides necessary hardware and software to define system status and alert

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maintenance personnel of actual failure or impending failure. Fault isolation down to the lowest possible level shall be provided. System alarms and fault detection error messages shall be routed to the interactive terminal where the system status can be visually displayed on the screen. As each alarm or error message is sent to the terminal, an audio alarm will be enabled at the maintenance position. This alarm remains enabled until maintenance personnel acknowledge all system alarms or error messages. Provision shall be made for remoting the alarm to the supervisor position. The maintenance terminal printer shall have the capability to output the following information as real-time (automatic). In addition, the status summary (see 3.3.2.5.6) shall be printed on demand.

3.3.2.5.1 Error messages. Error messages for failures, including power supply failure and low voltage alarms, shall be printed. These messages shall include time of failure, unit name, and type of failure. Once corrected or rebooted, a printout shall be made showing the time the unit went back in service.

3.3.2.5.2 Reconfiguration messages. Messages for any local or global map assignment shall be printed. Messages for local reconfiguration shall include the affected position, the time of reconfiguration, number of map assigned, and password number of who initiated the change. Messages for global reconfiguration shall include the time of reconfiguration and the password number.

3.3.2.5.3 Map change messages. Change messages for any modification to an existing map or the creation of a new one shall be printed. The map number and password number of the person doing the change shall be printed.

3.3.2.5.4 Diagnostic messages. Messages printed for any position off-line in diagnostics shall include time, position number, and the time the position returned to service.

3.3.2.5.5 Device initialization messages. Device initialization messages shall be printed with unit name and time included.

3.3.2.5.6 Status summary dump. A status summary dump of the system activity, i.e., a summary of unit failures, reboots, and reconfigurations, shall be a condensed report automatically printed every 4 hours or dumped on demand.

3.3.2.6 Maintenance control equipment subsystem. This subsystem provides maintenance personnel with necessary equipment to gain access to all radio, intercom and telephone channels through the utilization of a multi-page touchscreen entry/display. The specific requirements for the maintenance position shall be as specified in 3.3.2.6.1 through 3.3.2.6.5.

3.3.2.6.1 Multi-page touchscreen entry/display module (MPTED). This module shall be identical with the MPTED described in 3.3.2.4.1.

3.3.2.6.2 Speaker control module (SCM). This module shall be identical with the SCM described in 3.3.2.1.5.

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3.3.2.6.3 Jackbox module. This module shall provide the interface between the system and the designated operator using the supervisor position. It shall contain one set of tip-ring-sleeve jacks for use with 6-wire headsets/handsets equipped with PJ-421 plugs. The jacks shall occupy minimal space in a front-panel mounted module and shall be clearly designated for use by the operator.

3.3.2.6.4 Maintenance position processor module (MPPM). This module shall be identical with the OPPM described in 3.3.2.1.10 except for any changes necessary to provide additional channel accessing capabilities.

3.3.2.6.5 Handset. A 6-wire (W.E. G5JRW-3 or equivalent) shall be provided with each maintenance position.

3.3.2.6.6 Radio channel loop tester. This module shall provide maintenance personnel with the capability to perform a "quick" loop test from the maintenance position on each individual radio channel (transmit, receive, PTT). The module shall perform the same function as the TS-1505A (see NAVELEX 0967-874-5010). The TS-1505A is a fully portable device (powered by mike current), containing an audio oscillator (1KHz), a 6-wire headset plug, a front panel mounted meter for measuring the oscillator output and receive audio level, an oscillator level control and a switch for selecting between the internal oscillator and receive audio. A switch shall also be included to provide PTT capability. Provisions shall be made for defeating the receive radio mute feature at the maintenance position when conducting the loop tests. This will allow the tester to measure true sidetone, thereby, providing a complete loop test through the IVCSS transmit circuitry (audio and PTT), remote transmitter, remote receiver and IVCSS radio receive circuitry.

3.3.2.7 Power equipment subsystem. The IVCSS shall be equipped with a redundant DC power supply system that shall serve as the primary regulated DC power source. All supplies shall operate from single-phase, 115/230VAC + 10 percent, 47 to 63 Hz, input power. Complete battery back-up capability shall be provided which will allow the IVCSS to operate in a normal manner for up to 2 hours upon the failure of primary power source. Battery type and capacity will be specified by the contractor. The Government will supply the batteries at time of installation. Changeover to battery back-up will be automatic with no intervention required by operations or maintenance personnel. Alarm signals shall be provided for low voltage, high voltage, low charging current. These alarm signals shall be reported through the maintenance diagnostic equipment subsystem. The back-up capability may be provided by either of the two methods specified in 3.3.2.7.1 and 3.3.2.7.2.

3.3.2.7.1 Uninterruptible power system (UPS). An UPS is comprised of a battery charger, inverter, battery, battery racks and control circuitry. In normal operation, the battery charger maintains the batteries in a float condition and supplies DC power to the inverters. The inverter supplies 50/60Hz power to the IVCSS DC power supplies. The output of the inverter is synchronized to the incoming AC line. In the event of primary power failure, the batteries supply power to the inverter instantaneously. Thus, no interruption occurs in power delivered to the IVCSS. The UPS method facilitates the use of multi-voltage DC power supplies. Load current and voltage meters shall be provided on all main supplies.

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3.3.2.7.2 Single voltage source rectifier/charger. The single voltage source shall be comprised of a rectifier/charger, battery, battery rack, battery eliminator filter and control circuitry. In normal operation, the single voltage source rectifier/charger(s) maintain the batteries in a float condition and supplies power to the IVCSS equipment. In the event of primary power failure (or rectifier/charger failure) the batteries supply power directly to the IVCSS instantaneously. Thus, no interruption occurs in power delivered to the IVCSS. A battery eliminator filter provides necessary filtering of the rectifier/charger DC output if batteries are removed. Provision shall be made for disconnecting batteries for maintenance purposes. Load current and voltage meters shall be included on each rectifier/charger.

3.3.2.8 Traffic data collection (TDC). The IVCSS shall be capable of providing traffic data collection (TDC) so that the traffic load distribution within the system can be collected and analyzed on an off-line basis. The TDC capability may be integrated into the maintenance/supervisor positions or may be supplied as a portable unit capable of being transported from one site to another. If a portable unit is provided, each IVCSS shall be capable of interfacing with the portable TDC equipment. In addition, the portable TDC equipment shall be capable of being transported by one person. The TDC process shall be initiated and terminated by operator commands at the maintenance/supervisor positions or at the portable TDC equipment. The data shall include the following parameters for each position. Hardcopy printout of the data shall be provided.

a. Time of day, in one second intervals

b. Call initiation

c. Call answer

d. Radio channel identification, status, main/standby and PTT

e. Answer-back (busy, ring)

f. Position identification

g. Telephone identification

3.3.3 System operation. The IVCSS shall provide rapid, reliable communications for its operators by means of radio, intercom (intrafacility) and telephone (interfacility). All ATC position equipment, switching network equipment, interface equipment, interconnection busses, configuration control equipment, maintenance/diagnostic support equipment and power system equipment, as specified herein and delineated in the contract schedule, shall be integrated and provided as an operating system. All communications on radio, intercom and telephone shall be non-blocking and all connections must be completed within the permissible through connection delay time. There shall be no limit to the number of circuits (i.e., radio, intercom, or telephone) that can be connected in the system concurrently. Error checking, including as a minimum the verification of parity, shall be provided for transmission of data across all major interfaces (between distinct, separable and significant elements) in the system. Data errors shall be corrected, so that anomalous data does not result in system dysfunction. System faults

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shall be indicated by means of visual and audible alarms at a designated maintenance terminal. The operator position equipment described in this specification shall provide to operators various capabilities depending upon the configurations of both the positions and the system. This section describes the procedures and features which shall be provided in the course of initiating, receiving, conducting and terminating conversations in each of the three system modes (radio, intercom and telephone).

3.3.3.1 Radio operation

3.3.3.1.1 Radio channel selection. The system shall provide the capability of selecting or deselecting specific radio channels through a single action by the operator. A visual indication shall be given at the position when a frequency has been selected. All incoming audio on that radio channel shall be directed to every position which has selected the channel. Whenever a voice signal is present on a channel, a visible indication (VOX) shall be provided to each position which has been configured to receive that channel, regardless of the physical location of the position. The radio channel button shall be locking except for channels designated as emergency channels, which shall be non-locking. An operator may have control of a radio channel either when the channel is selected (select-to-lockout) or only when actually keying (key-to-lockout).

3.3.3.1.2 Main/standby selection. The system shall provide a means to individually select main or standby receivers and transmitters with selected receiver/transmitter status indicated at the main/standby transfer control module and at all positions equipped with the alternate RSM configuration described in 3.3.2.1.1. Selection of the main or standby transmitter and receiver shall be accomplished by means of locking pushbuttons.

3.3.3.1.3 Select-to-lockout mode. Only one operator position may select a radio channel which is configured for select-to-lockout. A visual channel-in-use (steady on) indication shall be provided to all programmed positions indicting that the channel has been selected. If selection is attempted at another position, both a visual and audible busy signal shall be provided to that operator. Operation of the PTT or footswitch shall mute the receive radio on all selected channels at the keying position only and sidetone shall be provided. In addition, the transmit status indicator associated with the transmit button for each selected radio channel shall flutter giving a positive indication that the channel is keyed at the radio interface.

3.3.3.1.4 Key-to-lockout mode. In the key-to-lockout mode, any programmed operator may select the same radio channel simultaneously. No position is locked out until one position operates his PTT or footswitch. At this time, a visual channel-in-use display shall be provided to all other programmed positions. In addition, transmission shall be inhibited on that channel for all other positions. Attempts to key the channel from any other position while the channel is already in use, shall result in both an audible and visual busy signal at that position. At the keying position, the transmit status indicator associated with the transmit button for each selected radio channel shall flutter giving a positive indication that the channel has been keyed at the radio interface and a VOX signal shall be received. In addition, all received radio audio shall be muted on all selected channels at that

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position and sidetone shall be provided.

3.3.3.1.5 Frequency sharing by remote radio operation. It shall be possible to interface a remote radio operator with the IVCSS, and share a local or remote transmitter and receiver. This operator shall hear the local controller's transmission as well as the radio receive audio. When the remote party keys, the local controller shall hear the remote's transmission.

3.3.3.1.6 Radio channel preemption. The supervisor and selected operator positions shall be able to transmit and receive with preemption capability on specific radio channels. Only one position will be programmed for preemption capability on a given radio channel.

a. Radio channels configured for select-to-lockout. Preempting positions may take control of any programmed radio channel by selecting the appropriate RSM transmit select pushbutton. This action transfers all radio operator functions except operator receiver monitoring from the preempted position to the preempting position for that particular channel. Both a visual and audible signal shall be provided to the preempted position indicating that preemption has taken place. The control of the channel shall return to the preempted position if released by the preempting position while the preempted position RSM transmit select pushbutton remains selected.

b. Radio channels configured for key-to-lockout. Upon activating the PTT or footswitch, a preempting position may take control of the radio channel being keyed by the preempted position. The preempted position shall hear a preemption tone indicating that the channel has been preempted. Receive radio audio to the preempted position shall not be affected by this action.

3.3.3.1.7 Received radio muting. The IVCSS shall be capable of radio receive muting on both a position and channel basis. Operation of the PTT or footswitch shall always mute received radio audio at the keying position. Under normal operation, all other programmed positions shall hear received radio audio at the normal receive level if the particular channel is selected at those positions. However, in some cases it is required to mute receive radio audio to all positions (on selective channels) whenever any position that has control of the channel operates the PTT or footswitch. Therefore, selective radio receive audio channel muting shall be electronically alterable.

3.3.3.2 Intercom operation. The IVCSS shall provide for both direct access (DA) and indirect access (IA) intercom with an option for override capabilities. In addition, a monitor only function shall be included. It shall be possible to program both a direct access intercom button and intercom monitor button to the same channel.

3.3.3.2.1 Direct access intercom. The DA pushbutton shall permit communication access to a designated position by a single operation of the DA pushbutton. A separate DA button is required for each location to be accessed. Each DA button may be configured for locking or non-locking operation.

a. Direct access intercom without override. Initiation of a direct access intercom call shall result in visual and aural signaling indicators at only one called position. At the called position, the appropriate button

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shall flash and either the chime shall sound or voice from the called position will be directed to the called position's speakers. At the calling position, the DA button light shall come on steady. At those called locations which have a corresponding direct access button, the call shall be answered by use of the appropriate DA button. At terminations which do not have a corresponding DA button, incoming calls shall be routed to the common answer (CA) button. If the CA button is already in use, the caller shall be placed in queue. (See 3.3.3.2.2.a.) After answering, the aural indicator shall turn off and the visual indicator shall flutter. Two-way conversation shall be enabled. The audio at each position shall be received according to the Headset/Loudspeaker (HL) pushbutton. Termination of a call made using a non-locking DA button shall occur as soon as the button is released. Either party may terminate a locking DA without override call by touching the DA button again, or by use of the release. When the call is terminated, indicator lights at both positions shall go out. The ability to terminate by selecting another DA button or the IA button shall be provided on an electronically alterable basis.

b. Direct access intercom with override. The call may be initiated either in locking or non-locking modes, as configured, in the same way as a direct access without override call. The visual indicators at both the calling and called position shall be the same as for a direct access without override. In addition, an override indicator (OHL lamp) shall flutter at the called position as long as the call is in progress. In case the OHL pushbutton has previously been selected to speaker, the OHL light shall not flutter but shall be on steady. No action shall be required on the part of the called position to receive the incoming override call. Zip tones shall be received at both the calling and called parties to indicate an override call has been connected. If the called position had a conversation in progress, the calling position enters the link in a conference mode and all parties shall be able to speak to and hear all others (including radio) with the exception that the overriding voice of the calling position shall not be transmitted over radio. Under these conditions, the zip tone shall be heard by all parties except that no zip tone should ever be transmitted over radio. Termination of an override call, by the called position, shall occur only after the IVCSS operator enters a unique code into the system. Termination of an override call made by means of a non-locking button at the calling position shall occur as soon as the button is released. Calls made by using a locking button shall be terminated by the calling party in the same manner as calls made without override. To ensure that vital communications at the called position is not interrupted, the calling party shall monitor all received audio at the called position. The received audio level shall not be affected by the setting of the called position's speaker or headphone level controls. Voice signals to the overridden position shall be directed to the headset, if the Override Headset/Loudspeaker (OHL) transfer button is in the "headset" position, or to the loudspeaker, if OHL is in the "loudspeaker" position. However, when the HL button is in "loudspeaker" position, all overrides shall be directed to the loudspeaker, regardless of the OHL button.

3.3.3.2.2 Indirect access intercom. The IA button in conjunction with the pushbutton dial keypad shall permit communication access to any position within the system. Indirect access shall be available regardless of whether the position to be called is accessible using a direct access button.

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a. Indirect access intercom without override. The call shall be initiated by use of the IA button followed by a two or three digit code specifying the particular local or remote position. The IA button at the calling position shall light when depressed and dial tone shall be heard in the operator's headset. The IA button shall be steady on until the call is answered. It shall then change from steady to flutter. Once the dial code is keyed in, the called positions chime shall sound (unless muted at the position) and the CA button shall flash. Should the called party possess a DA button corresponding to the caller, then the DA button (instead of the CA button) will flash. The called position will answer the call by depressing the flashing button (CA or DA). The flashing shall now change to flutter, the chime will cease and the connection shall be established. If the CA button was in use, the caller will encounter a busy condition and will be held in queue for answering by the operator in the sequence in which such calls arrive. While one call is in progress, three subsequent calls shall be held in this queue. The fifth and later calls receive busy signals. Answering of queued calls shall be accomplished by successive depressions of the CA button. The calling position shall be notified by a distinctive tone that he has been placed in queue. The CA button (light) shall indicate the status of the queue by using off, flutter, flash and wink conditons. Each party may terminate an indirect access call without override in the same manner as a direct access non-override call. In addition, the called party shall be able to terminate the connection if he has other calls in queue, by depressing the CA button.

b. Indirect access intercom with override. Override calls shall be initiated though the use of the appropriate two or three digit code number in the same manner as non-override indirect access calls. A position receiving such a call shall not be required to take any action to establish the connection. Termination of IA override calls by the called party shall be allowed only in the same manner as specified for DA override calls in 3.3.3.2.1.b.

3.3.3.2.3 Intercom monitor. Selection (locking or non-locking) of an intercom monitor pushbutton on the DA module shall cause the associated indicator light to come on steady at the monitoring position only. All audio at the position being monitored shall be received according to the intercom/telephone HL pushbutton. The received audio level shall not be affected by the setting of the monitored position's speaker or headphone level controls. Selection of the intercom monitor pushbutton will not cause the monitoring or monitored position's microphone circuit to be made "hot". If configured for non-locking operation, monitoring is terminated by releasing the DA button. When configured for locking operation, the operator must touch the DA button again in order to terminate monitoring. Upon initiation of a DA or IA call, received monitor audio is muted.

3.3.3.2.4 Loop conference net. When used in a range application, the IVCSS shall provide up to four loop conference nets. These nets shall provide high priority voice communications between multiple positions within the IVCSS and remote positions or remote switching systems (furnished by others). The IVCSS shall provide the capability to preprogram up to fifteen IVCSS operator positions to each of these nets via reconfiguration maps. Each programmed position shall have the capability of entering each net via two DA buttons. One DA button will provide monitor only capability while the second DA button

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provides the capability to communicate with all other positions on the net. Each net shall connect to the remote positions/remote switching systems via a 4-wire voice circuit.

3.3.3.3 Telephone operation. The IVCSS shall provide for:

a. Direct access telephone without override. Initiation of a DA telephone call shall cause the associated indicator light to come on steady green at the calling position and steady amber at all positions which have the same DA line. These indicators shall remain on until the call is terminated. When the called party answers, two-way communications shall be enabled. The audio at the calling position shall be received according to the Intercom/Telephone HL pushbutton. Direct access telephone communications shall be either locking or non-locking, as specified. When non-locking operation is specified, the call shall be terminated by releasing the DA button. Otherwise, the operator may terminate a call by touching the DA button again by selection of another intercom/telephone or by use of the release or indirect access button. Incoming telephone ringdown calls shall cause visual signaling indications at multiple positions which are configured to share the appearance of the circuit. The DA button at each configured position shall flash. The chime shall sound at each position which has been designated by the active configuration map to receive the ring for this circuit (unless muted at the position). Any of the positions having the circuit shall be able to answer the call. When one party answers the call, the chime shall stop at all positions. The light at the answering position shall go on steady green, and at all configured positions they shall go on steady amber to indicate the circuit is in use. Two-way communications shall be enabled. Incoming telephone voice calls shall cause the incoming voice of the calling party to be heard in the speakers of all positions which are configured to share the voice call circuit and which have been designated by the active configuration map to receive the voice audio. The DA button at each configured position shall flash. When the voice call is answered, the audio shall then be connected only to the answering position. The visual indicators shall be the same as for telephone ringdown calls. Initiating of calls requiring dial signaling shall be accomplished by accessing the appropriate DA button and then entering a two or three digit code on the SFM specifying the particular position at the remote facility. Only the called position shall receive signaling.

b. Direct access telephone with override. When remote facilities are equipped with systems capable of initiating and receiving telephone override calls, then the IVCSS shall provide for receiving and initiating such calls with those facilities. Telephone override calls shall have the same characteristics as intercom override calls.

3.3.3.4 Simultaneous operation. Activation of the PTT or footswitch at an operator position shall allow outgoing communications over selected radio channels while an intercom or telephone call is in progress. Under this condition, received intercom or telephone audio shall not be transmitted on any radio channel. Intercom or telephone communications in progress shall not be disturbed by PTT or footswitch activation except for the called party being able to hear the operator's transmission on radio. Received radio audio is transferred to the speaker upon selection of a telephone circuit.

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3.3.3.5 Headset/handset removal. Removal of all headsets/handsets from a jackbox module shall automatically disable all position pushbutton controls and transfer all received audio to the position speaker.

3.3.3.6 Call forwarding. Every programmed position shall be able to forward incoming intercom and telephone calls to another position. If the answering position is programmed for call forwarding, the assigned SFM or DA pushbutton shall be used for answering. If not, the CA pushbutton shall be used. Call forwarding shall be initiated and cancelled by entering appropriate codes on the SFM's or touchscreen's pushbutton dial keypad.

3.3.3.7 Conference calls. A conference call shall be initiated by first selecting the "conference" pushbutton and then sequentially selecting desired DA intercom pushbuttons or sequentially dialing appropriate codes via IA intercom.

3.3.4 Performance

3.3.4.1 System delays. The following system delay requirements shall be met.

3.3.4.1.1 System start-up. Normal system operation shall begin within 1 minute after the application of power to the system.

3.3.4.1.2 Push-to-talk (PTT) switching time. The time interval from actuation of the position PTT or footswitch until a keying signal appears at the system radio interface shall be 50 milliseconds (ms) or less.

3.3.4.1.3 Intercom DA and IA calls (non-override). The time interval from the activation of a DA pushbutton or an IA pushbutton sequence to the actuation of the chime and the audio connect at the called position shall be less than 300 ms.

3.3.4.1.4 Intercom DA and IA calls (override). The time interval from the activation of a DA pushbutton or an IA pushbutton sequence until audio connection is established and zip tone begins shall be less than 500 ms.

3.3.4.1.5 Telephone calls. The time interval from the activation of a DA pushbutton until the position is connected to a telephone circuit shall be less than 500 ms.

3.3.4.1.6 Dial tone. A dial tone shall be provided to a position within a period of 100 ms after activation of the IA pushbutton.

3.3.4.1.7 Call hold. An operator shall be able to place a call on hold within 200 ms after the function has been activated.

3.3.4.1.8 Call transfer. The call transfer feature shall be accomplished within 200 ms after its selection.

3.3.4.1.9 Call forward. The call forward feature shall become active within 200 ms after its selection.

3.3.4.1.10 Conference calls. A position shall be able to enter or

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establish a conference call within 200 ms after the function has been selected.

3.3.4.1.11 Transmitter and receiver main/standby selection. The time interval from the activation of the main/standby selection switch until a switching signal appears at the system radio interface shall be less than 300 ms for any selected radio channel.

3.3.4.1.12 Call and feature deselection. The deactivation of all functions shall occur within 200 ms after deactivation has been initiated. Function deactivation includes the elimination of all visual signals, audible signals and audio connects associated with each function.

3.3.4.2 Voice frequency circuits. The following voice frequency characteristics apply to the IVCSS and are valid up to the demarcation points associated with the various system interfaces.

3.3.4.2.1 Impedance. Each voice frequency circuit shall present a 600 ohm impedance, balanced to ground, to its interface with a leased or Government owned circuit at the point of demarcation.

3.3.4.2.2 Noise. The combined hum and noise of any idle voice frequency circuit measured at the position headset/handset jacks or voice path interfaces with both ends of the path properly terminated shall be not greater than -50 dBrnC.

3.3.4.2.3 Impulse noise. The peak level of impulse-type noise generated within the system when measuring on a single path of an idle voice frequency circuit, with both ends properly terminated, shall be not greater than three bits above a level of -50 dBrnC in a three minute period.

3.3.4.2.4 Harmonic distortion. Total harmonic distortion in a voice frequency circuit resulting from any single frequency between 300 Hz to 3000 Hz when applied at a level of -3 dBm0 shall be not greater than 2.5 percent.

3.3.4.2.5 Intermodulation distortion. The rms sum of all intermodulation products resulting from any two frequencies between 300 and 3000 Hz when applied to a VF circuit at a level of -3 dBm0 shall be not greater than -34 dBm0.

3.3.4.2.6 Frequency response. The voice signal amplitude at frequencies between 300 Hz to 3000 Hz shall be within $\pm$ 2 db of the 1000 Hz ($\pm$ 25 Hz) amplitude level. Roll-off characteristics of frequencies below 300 Hz and above 3000 Hz shall be at a minimum of 6 db per active. Additionally, the attenuation at 60 Hz shall be at -20 db down from nominal.

3.3.4.2.7 Crosstalk and isolation. The crosstalk between any two voice frequency transmission paths within the IVCSS shall be not less than -60 dB. The radio, intercom and telephone signals shall be isolated from each other by at least 60 dB. The above crosstalk and isolation figures represent what is required for unclassified communications. However, greater crosstalk and isolation (approaching 90 dB) would be desirable since it would facilitate the incorporation of a classified communications capability at some future date.

3.3.4.2.8 Longitudinal balance. The minimum longitudinal balance is 50

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dB over the frequency range from 300 Hz to 3000 Hz.

3.3.4.2.9 Return loss. A return loss of not less than 26 dB over the frequency range from 300 Hz to 3000 Hz shall be provided.

3.3.5 IVCSS voice level requirements. A broad range of voice levels are normally encountered on radio and telephone circuits terminated in Air Traffic Control Communications Systems at Navy facilities. Therefore, the various IVCSS interface circuits shall be adjustable to provide necessary compatibility. In addition, the IVCSS control central and position equipments shall provide voice level regulation to the degree specified herein. Surge protection shall be provided at the jackbox module to prevent any signal exceeding a peak value of -8dBm from reaching the headset or handset.

3.3.5.1 Received radio levels. The nominal radio receive level at Navy facilities is 0dBm. However, each radio receive channel interface shall be capable of compensating for a variable input level range of -33dBm to +3dBm for both main and standby channels and still provide the specified headset nominal level at the jackbox module. The receive amplifier shall be capable of driving all operator positions simultaneously without any appreciable change in receive level at the operator positions. With a receive signal input which complies with the range specified above, the position's headset volume control shall provide for a level range of -45dBm to -10dBm at the jackbox module.

3.3.5.2 Received landline levels. The nominal telephone receive level at Navy facilities on 4 wire circuits is -15dBm. However, each telephone channel interface shall be capable of compensating for a variable input level range of -40dBm to 0dBm and still provide the specified headset nominal level at the jackbox with up to a ±8db change in the received level of a -8dBm telephone test tone signal, the received output signal at the position shall remain within ±1.5db of the nominal receive level. The receive amplifier shall be capable of driving all operators without any appreciable change in receive level at the operator position. With a receive signal input which complies with the range specified above, the position headset volume control shall provide for a range of -45dBm to -10dBm at the jackbox.

3.3.5.3 Transmit levels. The nominal transmit level at the Navy demarcation point is 0dBm. However, each radio and telephone channel interface shall be capable of providing an output which is adjustable from -10dBm to +3dBm. the system shall be capable of providing automatic voice level regulation in all transmitting voice paths from any position to maintain the output signal level within ±2.0 dB. The level regulation shall be maintained over a range of ±6 dB from the test tone level, which is defined as -15 dBm into a 600 ohm to 50 ohm line matching transformer at the tip contacts of the jackbox module headset jacks. A 12 dB sudden increase or decrease shall affect the output level in accordance with the following requirements.

a. 12 dB sudden increase (-6 dB below to +6 dB above the test tone). The instantaneous output level, including transients, shall not increase by more than 5 dB or decrease by more than 3 dB. The output level shall be within ±0.5 dB of the final steady state value within 10ms from the instant of input level change.

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b. 12 dB sudden decrease (+6 dB above to -6 dB below the test tone). Immediately following the 10ms stabilization period after the 12 dB increase, and with a sudden 12 dB decrease, the output shall stabilize to within 2 dB of the final steady state value in not less than 400 and no more than 600ms from the instant of input level change.

c. Squelch. Squelch circuitry shall be included in the microphone amplifier. A variable threshold adjustment (technician adjusted) shall provide the means to set the minimum voice signal level at which the amplifier turns on. The attack time and decay times shall be in consonance with the times specified in 3.3.5.3.a. and 3.3.5.3.b. The squelch feature is necessary to ensure proper system operation when operator positions are located in noisy environments.

3.3.5.4 Intercom levels. With up to a ±8 dB change in the received level of an intercom test tone signal, the received signal shall remain within ±1.5 dB of the nominal receive level.

3.3.5.5 Voice recording levels. All incoming or outgoing radio, intercom and telephone communications shall be combined into a single signal for recording. The nominal level supplied to the recorder shall be 0 dBm with the following position amplifier levels:

a.	Microphone	-15 dBm into 600/50 ohms line matching transformer
b.	Headset	-28 dBm
c.	Speaker	-10 dBm

3.3.5.6 Sidetone levels. Sidetone at the sleeve contacts of the jackbox module shall be 16 dB ±2 dB below the transmit signal present at the tip contacts of the jackbox. Position level control settings shall not affect sidetone levels. When a position is transmitting, the sidetone leakage across the speaker terminals shall be not less than 50 dB below the received audio level set by the speaker level control. Sidetone level shall not be affected by multiple accessing of radio, intercom and telephone channels.

3.3.6 Information tones. The information tones generated by the system shall be in accordance with Table II. The power per frequency specified in the table shall be measured at the headset jack.

3.3.7 DTMF pushbutton signaling tones. Pushbutton signaling tones shall be in accordance with Table III.

3.3.8 Repetitive tone suppression. It is desirable that extraneous repetitive tones or constant frequency noise signals in the voice band not be generated or propagated by the system.

3.3.9 Visual indication signals. All visual indication signals shall be in accordance with Table IV.

3.4 Computer firmware/software. Computer firmware and software development shall comply with DoD-STD-2167 to the extent specified in the statement of work (SOW).

3.5 Parts, materials and processes. Parts, materials and processes shall

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be as specified in 3.5.1 through 3.5.7.

3.5.1 Parts selection. Only integrated circuits, transistors and diodes available as standard catalog items from two or more manufacturers shall be employed. These devices shall meet the requirements of those commonly referred to as commercial specification. Plastic encapsulated devices may be employed. Instruction books or equipment manuals shall include a programming list, with sufficient explanation, for all read only memory (ROM) devices.

3.5.2 Parts availability. The parts or material used shall be available in the United States of America.

3.5.3 Parts quality. Selection of materials, processes, and parts shall be consistent with the requirement of producing a system that is not necessarily a completely MIL-Spec design type, but rather a system economically produced and capable of performing its specified functions with ruggedness and durability. This requirement shall be satisfied by the selection of industrially proven low cost materials, processes and parts that do not degrade the quality of the system.

3.5.4 Recycled material. All parts and materials used in the IVCSS shall be new.

3.5.5 Toxicity. The material chosen shall be of low toxicity not having dangerous gasses due to fire or toxic effects when used in a normal manner.

3.5.6 Fungus. The materials chosen shall be non-nutrient to fungus and insects, flame resistant, non-hydroscopic, and not adversely affected by the environmental conditions specified herein before.

3.5.7 Plastic Parts. Plastics used in conjunction with displays, pushbutton selectors, back lighting or other parts involved with illumination transmission shall not warp, deform, discolor or change in mechanical appearance or physical properties because of the concentration of heat product resulting from the illuminations.

3.5.8 Soldering and soldering processes. Soldering and soldering procedures and processes shall conform to WS-6536, except for commercial off-the-shelf items.

3.6 Environmental conditions. The IVCSS shall operate properly within a through f:

a.	Ambient temperature	Range 4 of MIL-E-16400
b.	Relative humidity	80% (±5 percent)
c.	Altitude	0 ft to 10,000 ft above mean sea level
d.	Ambient temperature (non-operating or storage)	-62 to +71°C
e.	Relative humidity (non-operating or storage)	95% (±5%)
f.	Altitude (non-operating or storage)	0 ft to 50,000 ft above mean sea level

3.7 Design and construction. Design and construction shall be as specified in 3.7.1 through 3.7.14.

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3.7.1 Cabling. The contractor shall provide all necessary intraconnecting cables and connectors for voice, control and power at both control central and each position subsystem. Interconnecting cable type between control central and each position subsystem shall be specified by the contractor and supplied at time of installation as GFM. Each interconnecting cable shall be terminated by the Government in contractor supplied connectors.

3.7.2 Position subsystem equipment. The physical dimensions of position equipment shall conform to the limitations depicted on Figure 2. All position modules shall be assembled (fastened together) and be ready for mounting in existing cabinet space. It should be noted that Figure 2 illustrates available space in an ASR position only. It is the contractor's responsibility to determine specific requirements of each position at each site. All pushbuttons (with the exception of the MPTED) shall provide tactile feedback. The MPTED pushbutton facsimile shall provide an audible feedback.

3.7.3 Cabinets construction. Control central cabinets shall be not greater than 84 inches in height, 26 inches in width and 26 inches in depth. The cabinets shall not apply an average weight distribution or floor loading greater than 175 lb/ft^2. Panel spacing shall comply with EIA RS-310-C-77.

3.7.4 Modularity. The IVCSS shall be modular in construction so that operator positions, radio, intercom and telephone channels may be added or removed incrementally without disruption to system operation. Expansion shall be accomplished by the addition of printed circuit cards, plug-in modules, cabinet subassemblies and cabinets.

3.7.5 Interchangeability. All components, assemblies, subassemblies and modules that are identical with respect to fit, form and function shall be interchangeable.

3.7.6 Finishes. All control central equipment cabinets shall be fully painted on all interior and exterior metal surfaces. All equipment exterior surfaces shall be painted in accordance with appropriate paragraph (painting, protected equipment, interior use) of MIL-E-16400 except for the color which shall be gray no. 26307 in accordance with FED-STD-595.

3.7.7 Rough handling. The IVCSS shall be sufficiently rugged, electrically and mechanically to withstand the degree of shock and vibration normally encountered during shipping, storage, and maintenance.

3.7.8 Power source. The IVCSS shall operate within the requirements of this specification when connected to a source of single-phase, 115/230 VAC ±10 percent, 50 Hz or 60Hz (nominal) power. Overvoltage and short circuit protection shall be incorporated into all power supplies. Transient and spike voltage protection shall also be included.

3.7.9 Bonding and grounding. The IVCSS shall conform to MIL-E-16400.

3.7.10 Indicators. All status lights and displays with the exception of CRTs, shall be readily visible when viewed in high ambient sunlight such as found in ATC control towers. SDMs shall utilize light emitting diode, liquid crystal or plasma displays which are viewable from sufficiently wide viewing angles. All types of indicator-displays shall be readily replaceable by

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station personnel. Incandescent lamps, where used, shall be Type T-1, three-quarters miniature flange base in accordance with MIL-L-6363 for 20,000 hour rated life. Use of other types of incandescent lamps must be approved by the procuring activity.

3.7.11 Safety requirements. The IVCSS design shall conform to MIL-E-16400.

3.7.12 Electromagnetic compatibility (EMC). The equipment shall be self-compatible and shall operate in the electromagnetic environment in which it is intended to operate. The equipment shall conform to the requirements of MIL-STD-461 for Class A3 equipment. MIL-HDBK-241 may be used for power supply design guidance.

3.7.13 Marking and labeling. Component and function identifications shall be in accordance with MIL-E-16400. Silk screening shall be used for all front and rear panel markings. PCB marking shall be silk screened or similar permanent method.

3.7.14 Cooling. All of the equipment shall use simple cooling techniques based on conduction, radiation, and free convection, using room air, to the maximum extent possible. Equipment requiring separate dedicated cooling systems shall not be used. Forced air cooling shall be used only when free air cooling is inadequate. All equipment shall operate whether the doors are open or closed. No forced air cooling shall be used in operator position equipment.

3.7.15 Channel/position-isolation. The system shall be designed to eliminate interaction between powered-down channels/position equipment and active channel/positions. That is, if power is removed or equipment modules are removed from a position or channel, other active channels or positions shall not be affected in any manner.

3.8 Quantitative reliability requirements. The lower test MTBF of the IVCSS, including software failure, (O_1 as defined by MIL-STD-781) shall be 500 hours.

3.9 Quantitative maintainability requirements. The equipment shall have a specified mean-corrective-maintenance-time of not more than 0.25 hour when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, PCBs) and chassis-mounted components (this includes electronic, electrical, electromechanical and mechanical parts). The equipment corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions.

3.10 Workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause of suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c. Safety verification (see 4.5.7)
d. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one IVCSS shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE V.

4.4. Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.5.

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4.4.1 Production inspection (Group A). Production inspection shall be performed on every IVCSS offered for delivery. Production inspection shall comprise such examination and testing that will prove the workmanship and reveal omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the examinations and tests specified in Group A of TABLE V.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.4. Production control inspection shall consist of examinations and tests which encompass functional and performance tests throughout the entire range of operation; and tests to determine deviations in the processes and equipment required to fabricate the equipment. These tests shall be performed on the complete equipment as offerer for delivery. Production control inspection shall include the examinations and tests specified in TABLE V.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.4. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and equipment. Environmental inspections shall include the examinations and tests in Group C of TABLE V. For tests requiring a chamber, the IVCSS shall be divided into its subgroups for purposes of environmental tests, subject to the approval of the procuring activity.

4.4.4 Sampling plan for production control inspection (Group B) and environmental inspection (Group C). One sample shall be selected from each, or fraction thereof, produced. The first sample shall be selected from the first month's production.

4.4.4.1 Nonconforming sample units. If sample units fail an inspection, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the IVCSS to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the procuring activity.

4.4.5 Reinspection of conforming production control and environmental sample units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to, and have passed, production control and environmental inspections may be accepted on the contract provided they are resubjected to, and have passed production control and environmental inspection, after repair of all visible damage.

4.5 Test methods. The examination and tests specified in TABLE V shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.1 through 4.5.9.

4.5.1 Surface examination. The IVCSS units shall be examined to ensure compliance with the requirements specified in a through e:

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a. Workmanship during fabrication of all items
b. Examination for marking
c. Examination of finish
d. Examination of parts and materials
e. Examination for ease of operation of mechanical assemblies

4.5.2 Software acceptance testing. Software acceptance testing shall be conducted in accordance with the software acceptance procedures stated in the SOW, and Government-approved test plans, test specifications, and test procedures to ensure conformance with formal testing requirements.

4.5.3 Soldering. Examinations shall be performed to verify conformance to the requirements of 3.5.8.

4.5.4 Design and construction test. A complete equipment operating test (see 4.5.4.1 and 4.5.4.2) shall be conducted on the IVCSS.

4.5.4.1 Unit operation test. Unit operational testing shall be conducted on each completed unit to verify compliance with unit functional requirements specified in 3.3.2.4.

4.5.4.2 Operational system test. Each IVCSS shall be assembled and tested as a system at the contractor's facility. The system test shall demonstrate conformance to all system and functional operations specified in 3.3.

4.5.5 Rough handling tests. Rough handling tests shall be conducted in accordance with MIL-P-116.

4.5.6 Performance. The IVCSS shall be subjected to the tests specified in 4.5.4.1 and 4.5.4.2 to verify conformance to 3.3.

4.5.7 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.7.11.

4.5.8 Leakage current test. Leakage current shall be measured at the steady-state power line voltages and frequencies of 115 volts (V).

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE VISCOM DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.9 EMC tests. The IVCSS shall be tested to verify conformance to the requirements of 3.7.12, using the test methods of MIL-STD-462.

4.5.10 Power source tests. The IVCSS power sources shall be tested to verify conformance to the requirements 3.3.2.7 and 3.7.8, using the electrical

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test methods of MIL-E-16400.

4.6 Reliability qualification test (RQT). The RQT shall be performed in accordance with MIL-STD-781 on not less than one production IVCSS. Testing shall be performed in accordance with MIL-STD-781, Test Plan IVC, and shall proceed to a decision in accordance with the criteria specified in MIL-STD-781, Test Plan IVC. The RQT conditions shall be in accordance with MIL-STD-781, Equipment Category 1, and the fixed ground requirements specified in the test conditions matrix table.

4.6.1 Reliability testing. Reliability testing shall be performed on an IVCSS which has been subjected only to production inspection. Testing shall be performed at Test Level E with Test Plan XXVIII of MIL-STD-781. The IVCSS shall be subjected to 48 operating hours at a temperature in accordance with 3.6.

4.7 Maintainability demonstration. Maintainability demonstration shall be performed in accordance with MIL-STD-471. Accept or reject criteria shall be in accordance with Method 4 of MIL-STD-471. For maintainability demonstration purposes, the equipment-repair-time is equivalent to the mean-corrective-maintenance-time.

4.8 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The IVCSS is intended for use in providing air traffic control and monitor operations at various Naval shore sites.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Number of first article samples to be submitted if other than specified in 4.3.
c. When reinspected production control and environmental sample units may not be accepted (see 4.4.5).
d. Levels of preservation, packaging, packing, and marking (see 5.1).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
4.7	Maintainability Program Plan	DI-R-7103	--
4.6	Reliability Test Plan	DI-RELI-80250	--
3.8, 4.6.1	Reports, Reliability Test and Demonstration	DI-RELI-80252	--
3.9, 4.7	Report Maintainability Prediction	DI-R-7108	--
4.0	Quality Program Plan	DI-R-5297A	--
3.7.11	System Safety Program Plan	DI-SAFT-80100	--
3.7.11	System Safety Hazard Analysis Report	DI-SAFT-80101	--
4.5.2	Software Test Plan	DI-MCCR-80014	--
4.5.2	Software Test Report	DI-MCCR-80017	--
3.7.12	EMC Control Plan	DI-EMCS-80199	--
4.5.9	EMI/EMC Test Plan	DI-EMCS-80201	--
4.3	First Article Inspection Report	DI-T-4902	--

(Date item description related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Initial shipment. Initial shipment of end items will be contingent upon successful completion of the RQT.

6.5 Production control inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. Initial shipment of end items will be contingent upon successful completion of production control inspection.

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6.6 Abbreviations and definitions. An explanation of abbreviations and definitions used in this specification is given below:

AC	Alternating Current
APS	Administrative Phone System
ATC	Air Traffic Control
Automatic Ring	Signaling accomplished through one operator action (i.e. selection of a DA switch at an operator position or removal of the handset from the hookswitch at a remote position)
C	Centigrade
CA	Call Answer
CO	Central Office
Classmark	A program code that enables or disables access to IVCSS services and functions
Configuration Maps	A correspondence set between IVCSS hardware elements and software elements that provides communications assignments and capabilities through applications of operational and service classmarks
CRT	Cathode Ray Tube
DA	Direct Access
DASM	Direct Access Switch Module
dB	Decibel
dBm	Power in dB relative to one milliwatt
dBm0	The test tone 1000 Hz power level at the reference point
dBr	dB relative to the point of zero transmission level
dBrnC	The noise power measured at the reference point with a "C" message weighting network referenced to the reference noise power level of .001 microwatts at 1000 Hz
DC	Direct Current
DTMF	Dual Tone Multifrequency

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E&M Signaling	DC signaling interface to certain trunk circuits. Traditionally, the E&M interface consists of two leads between the switching equipment and the signaling equipment with the M lead carrying dc signals from the switching equipment and the E lead carrying dc signals to the switching equipment
EMC	Electromagnetic Compatibility
EPROM	Eraseable Programmable Read-Only Memory
FCC	Federal Communications Commission
Flash	60 impulse per minute visual signal, with a 50/50 on/off ratio
Flutter	720 impulses per minute visual signal, with a 95/5 on/off ratio
4 Wire Circuit	A circuit that provides two-way communication over two separate pairs of conductors. One pair is used solely for transmitting and the second is used solely for receiving.
Fully Manual Signaling	Continuous signaling accomplished through two operator actions (i.e., the operator must select a DA switch and depress the RING switch. Called position's bell rings continuously until RING switch is released.
GFE	Government Furnished Equipment
GFM	Government Furnished Material
HL	Headset/Loudspeaker
HLCM	Headset Level Control Module
Hz	Hertz
IA	Indirect Access
INST	Instructor
Intercom	Intrafacility Ground-Ground Communications (referred to as interphone in AN/FSA-52)
I/O	Input/Output
IPM	Interruptions per Minute
IR	Infrared

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
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IVCSS	Integrated Voice Communication Switching System
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LMSML	Local Multi-station Milti-link Circuit
Manually Initiated Signaling	Signaling accomplished through two operator actions (i.e., the operator must select a DA switch and momentarily depress the ring switch)
MF	Multifrequency
MPPM	Maintenance Position Processor Modules
MPTED	Multi-Page Touch-Screen Entry/Display Module
ms	Millisecond
MTBF	Mean time between failures
Off-the-Shelf	Equipment which has been designed and is demonstrable for sale in either domestic or foreign markets, and is readily available for delivery
OHL	Override Headset/Loudspeaker
OPPM	Operator Position Processor Module
PBX	Private Branch Exchange
PTT	Push-to-Talk
QAR	Quality Assurance representative
Radio Channel	A communications path, including transmit, receive, and PTT capability to remote or local radio transmitters and receivers on a particular frequency
REL	Release Switch
RF	Radio Frequency
rms	Root Mean Square
RQT	Reliability Qualification Test
RSM	Radiophone Switch Module

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SCM	Speaker Control Module
SDM	Selector Display Module
SF Signaling	2600 Hz Single Frequency in-band signaling
SFM	Special Function Module
SO	Split Operation
SPPM	Supervisor Position Processor Module
TCM	Transfer Control Module (Main/Standby)
TDC	Traffic Data Collection
Telephone	Interfacility Ground-Ground Communications (referred to as landline in OA-7621/FSA-52)
2 Wire Circuit	A circuit that provides two-way communication over a single pair of conductors
UHF	Ultra High Frequency
UPS	Uninterruptible Power System
VAC	Volts Alternating Current
VDC	Volts Direct Current
VF	Voice Frequency
VFL	Vacuum Fluorescent
VHF	Very High Frequency
VOX	Voice Operated Switch
Wink	60 impulses per minute visual signal, with a 95/5 on/off ratio
Zip Tone	A 200 ms burst of the mixed frequencies 350 Hz and 440 Hz

6.7 Subject term (key word) listing

Air traffic control

Central office

Configuration maps

Crosstalk

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DTMF Signaling

Instructor

Intercom

Jackbox

Microprocessor

Non-blocking

Operator

Override

Pre-empt

Push-to-talk

Radiophone

Ringdown

Selector

Sidetone

Supervisor

Switching system

Telephone

Touchscreen entry/display

Traffic data collection

Voice communication

Voice recording

Wink

Ziptone

Preparing Activity
Navy-AS
(Project 5895-N)

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Table I. Telephone circuit configurations

NAVY Line Type	No. Of Wires	Incoming Signal	Outgoing Signals	Required OA-7621 Termination Equip	Applicable Technical Manual(s)
1	2	Voice	Voice	AM-6177	0967-LP-005-9000
2	2	Ring (20 Hz)	Manual Ring (20 Hz)	AM-6177/CV-2601	0967-LP-005-9000
3	2	Voice	Manual Ring (20 Hz)	AM-6177/CV-2601	0967-LP-005-9000
4	2	Ring (20 Hz)	Voice	AM-6177/CV-2601	0967-LP-005-9000
5	4	Voice	Voice	AM-6177	0967-LP-005-9000
5A	4	Voice	Voice & Manual Ring	AM-6177	0967-LP-005-9000
6	4	Ring (20 Hz)	Manual Ring (20 Hz)	AM-6177/CV-2601	0967-LP-005-9000
6A	4	Ring Simplex (20 Hz)	Manual Ring (Simplex 20 Hz)	AM-6177/CV-2601	0967-LP-005-9000
6B	4	Ring (2600 Hz)	Manual Ring (2600 Hz)	AM-6177/Unit 87	0967-LP-005-9000
7	4	Ring (SS-1/SS-4)	Dial (SS-1/SS-4)	AM-6177/WESCOM/ CV-3654	0967-LP-005-9000 & LAB 1760
8	4	Voice	Manual Ring (20 Hz)	AM-6177/CV2601	0967-LP-005-9000
8A	4	Voice	Manual Ring (Simplex 20 Hz)	AM-6177/CV2601	0967-LP-005-9000

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Table I. Telephone circuit configurations - continued

NAVY Line Type	No. Of Wires	Incoming Signal	Outgoing Signals	Required OA-7621 Termination Equip	Applicable Technical Manual(s)
9	4	Ring (20 Hz)	Voice	AM-6177/CV-2601	0967-LP-005-9000
9A	4	Ring (Simplex 20 Hz)	Voice	AM-6177/CV-2601	0967-LP-005-9000
10	4	Voice	Dial (SS-1/Ss-4) (20 Hz)	AM-6177/CV-2911A	0967-LP-005-9010
11	4	Ring (SS-1/SS-4)	Voice	AM-6177/WESCOM	0967-LP-005-9000
12	2	Ring (20 Hz)	Dial (Loop Pulse)	AM-6177/CV-2892A	0967-LP-005-9020
16	2	Ring (20 Hz)	Auto Ring (Loop Closure)	AM-6177/CV-2892A	0967-LP-005-9020
18	2	Ring (DTMF or Automatic or Voice)	Ring Automatic or Manual)	AM-6177/CV-3613	0967-LP-005-9040

NOTE: Types 13, 14, 15 and 17 are not applicable.

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Table II. Information tones

Signal	Frequencies (Hz) (+0.5%)	Power Per Frequency (dBm0) (+1.5 dB)	Interruption Rate	Tone-On (Sec)	Tone-Off (Sec)
Dial Tone	350 + 440 (Mixed)	-16	Continuous	-	-
Line Busy	480 + 620 (Mixed)	-24	60 IPM	0.5	0.5
Trunk Busy	480 + 620 (Mixed)	-24	120 IPM	0.2	0.3
Reorder	480 + 620 (Mixed)	-24	120 IPM	0.2	0.3
Audible Ringing Routine	440 + 480 (Mixed)	-16	10 IPM	2.0	4.0
Audible Ringing Preference	440 + 480 (Mixed)	-16	30 IPM	1.64	0.36
Pre-Emption	440 + 620 (Mixed)	-18	Continuous	-	-
Conference Notification	852 + 1336 (Alternated)	-24	NOTE 1	-	-
Permanent	350 + 480 (Mixed)	-17	Continuous	-	-
Zip Tone	350 + 440 (Mixed)	-16	One Burst	0.2	-
Beep (NOTE 2)	1400	-16	Repeated	0.3	15 - 17
Chime	500, 750, 1000, 1250, 1500	NOTE 3	One impulse every 2 sec. with a following decay	-	-

Notes

NOTE 1: The tones shall alternate at 300 ms intervals.

NOTE 2: The beep tone shall be delayed at least 15 seconds after the completion of dialing. When so specified in the contract schedule, the beep tone requirement shall be deleted.

NOTE 3: See 3.3.2.1.5

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Table III. Pushbutton signaling tones

Digit	Pushbutton Designation	Tone Pair
Digit 1	1	697 and 1209
Digit 2	2	697 and 1336
Digit 3	3	697 and 1477
Digit 4	4	770 and 1209
Digit 5	5	770 and 1336
Digit 6	6	770 and 1477
Digit 7	7	852 and 1209
Digit 8	8	852 and 1336
Digit 9	9	852 and 1477
Digit 0	0	941 and 1336
Spare		941 and 1209
Spare		941 and 1477

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Table IV. Visual indication signals

Signal	Purpose	Interruption Rate	Signal On (Percent)	Signal Off (Percent)
FLASH	Incoming call being received and position has not answered	60 IPM	50	50
FLUTTER	Intercom connection completion, radio PTT received at radio interface, VOX indication	720 IPM	95	5
STEADY ON	Intercom call or special function initiated. Telephone Circuit in use, radio circuit selected	Continuous	-	-
WINK	Circuit placed in hold condition	60 IPM	95	5

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TABLE V. Examination and Tests

Subject	Requirement	Test Method	First Article	Quality Conformance Inspection		
				Group (A)	Group (B)	Group (C)
Surface examination						
Weight	3.7.3	-	X			
Size	3.7.2,3.7.3	-	X			
Parts & materials	3.5	4.5	X	X		
Finish	3.7.6	4.5	X	X		
Marking	3.7.13	4.5	X	X		
Safety	3.7.11	4.5.7	X	X		
Workmanship	3.10	4.5	X	X		
Rough Handling	3.7.7	4.5.5	X			
Performance	3.3	4.5.6	X	X		
Soldering	3.5.8	4.5.3	X			
Environmental	3.6	4.5	X			
Steady state voltage and frequency	3.7.8	4.5.10	X			
Transient voltage	3.7.8	4.5.10	X			
Spike Voltage	3.7.8	4.5.10	X			
Leakage current	3.7.11	4.5.8	X			
Electromagnetic Compatibility	3.7.12	4.5.9	X			
Reliability	3.8	4.6	X			
Maintainability	3.9	4.7	X			

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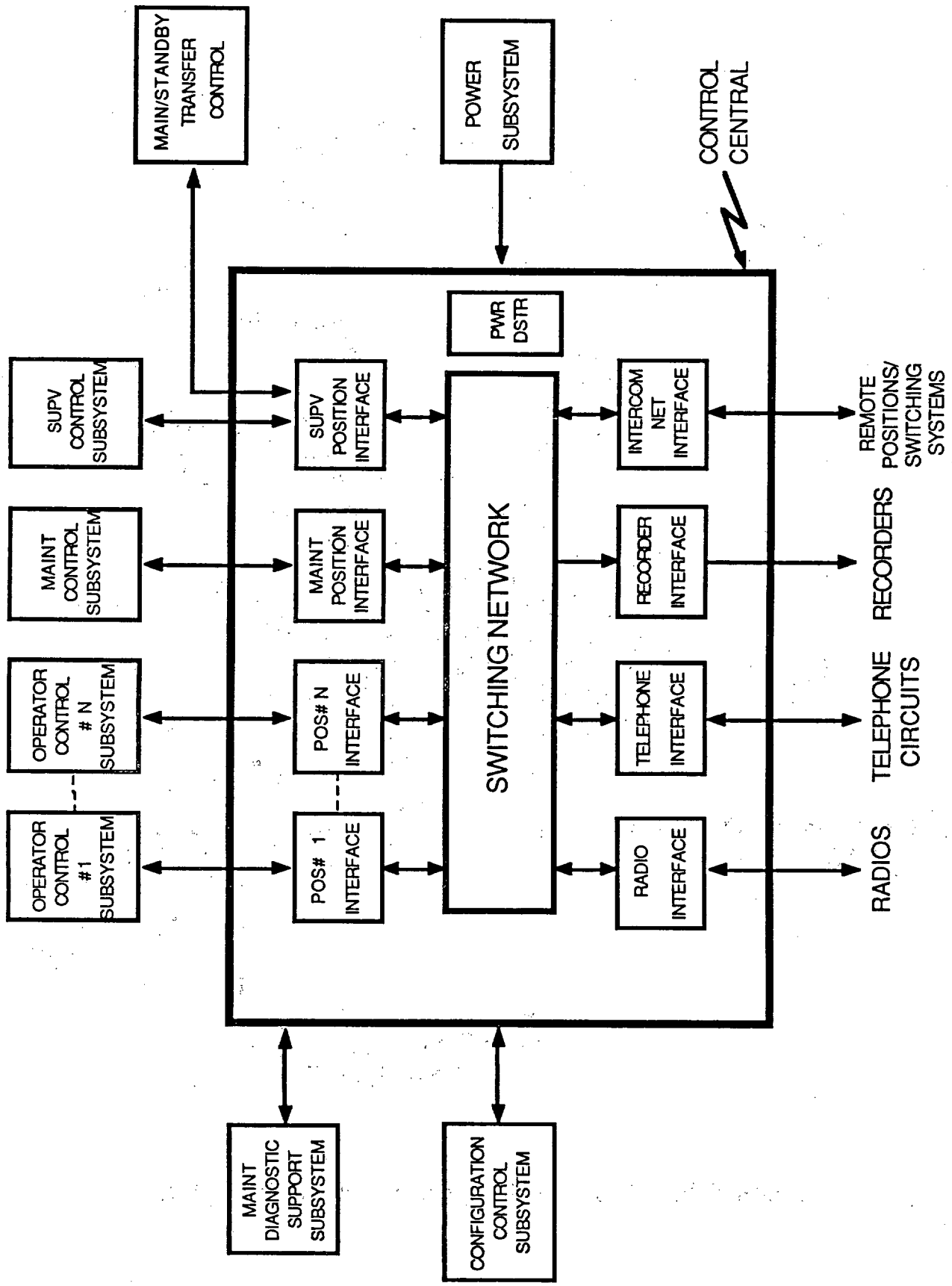


FIGURE 1. IVCSS functional block diagram

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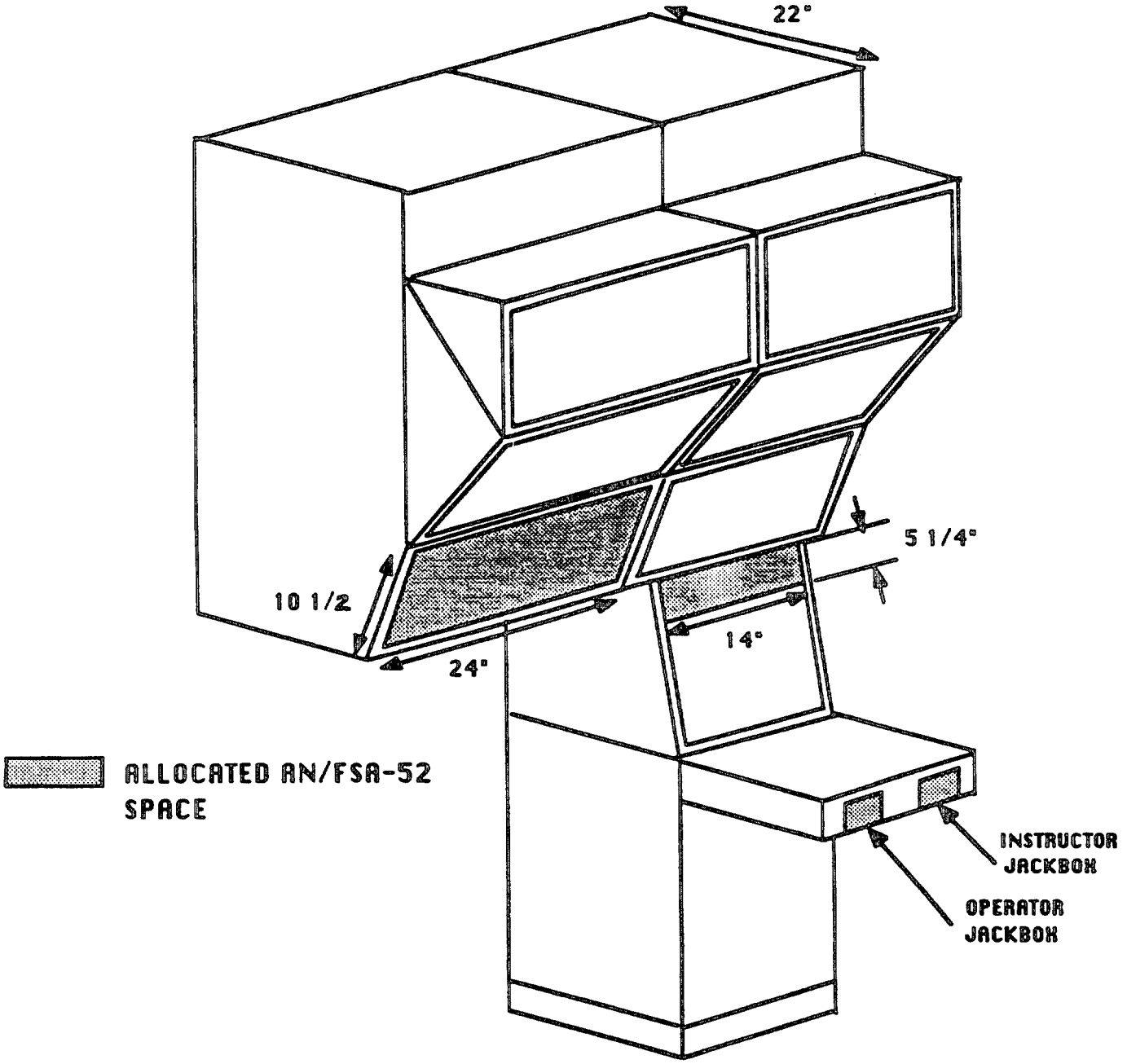


FIGURE 2. Typical ASR position for AN/FSA-52(V)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-85850	INTEGRATED VOICE COMMUNICATIONS SWITCHING SYS.

a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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